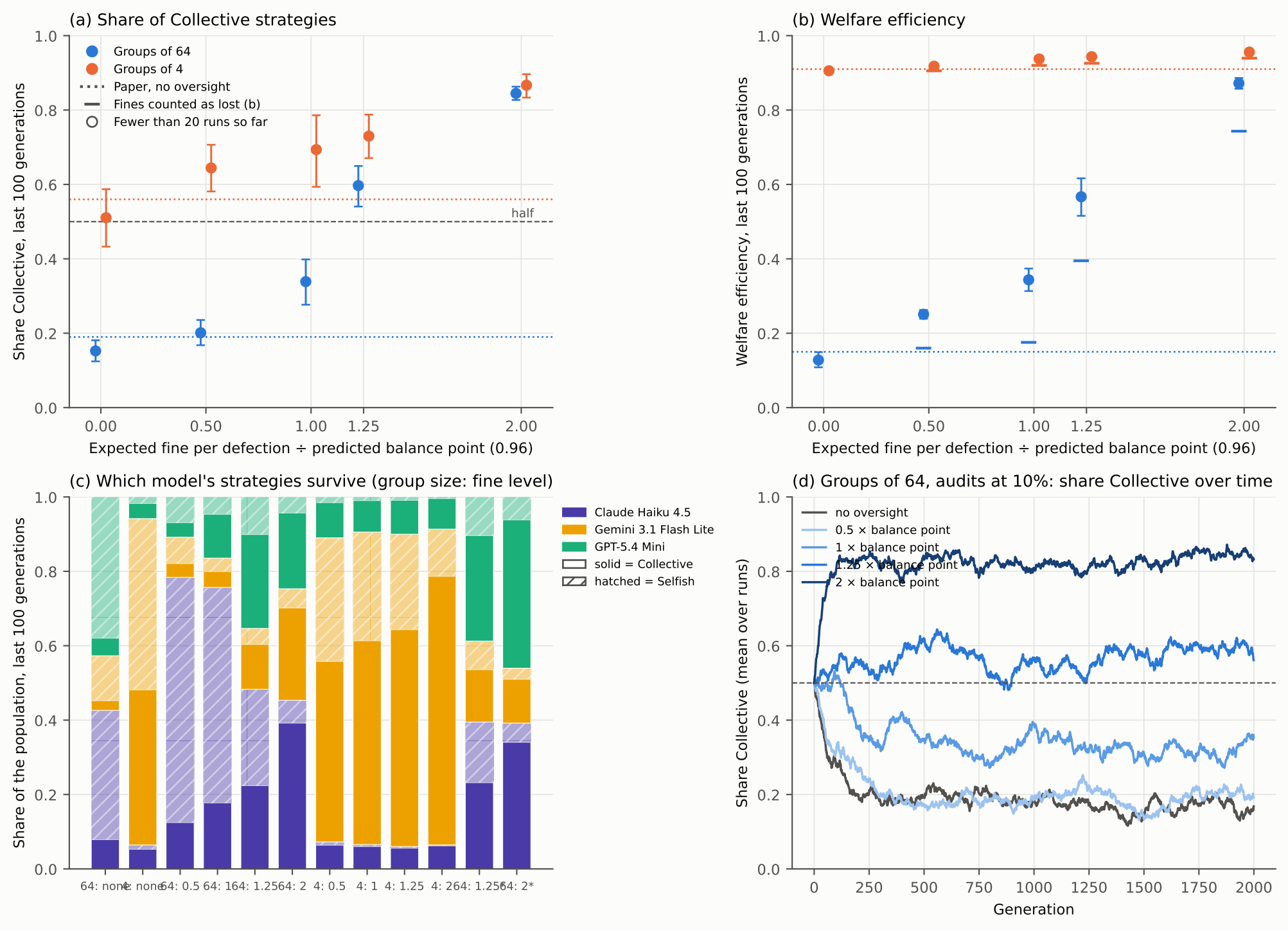

(a) Share of Collective strategies
Groups of 64
Groups of 4
Paper, no oversight
Fines counted as lost (b)
Fewer than 20 runs so far
half
Share Collective, last 100 generations
Expected fine per defection ÷ predicted balance point (0.96)
(b) Welfare efficiency
Welfare efficiency, last 100 generations
Expected fine per defection ÷ predicted balance point (0.96)
(c) Which model's strategies survive (group size: fine level)
Claude Haiku 4.5
Gemini 3.1 Flash Lite
GPT-5.4 Mini
solid = Collective
hatched = Selfish
Share of the population, last 100 generations
64: none
4: none
64: 0.5
64: 1
64: 1.25
64: 2
4: 0.5
4: 1
4: 1.25
4: 2
64: 1.25
64: 2*
(d) Groups of 64, audits at 10%: share Collective over time
no oversight
0.5 × balance point
1 × balance point
1.25 × balance point
2 × balance point
Share Collective (mean over runs)
Generation